

Neuronal Resuscitation with Stem Cells in Spinal Cord Injury : Clinical and Histopathological Evaluation in Umbilical Mesenchymal Stem Cells Delivered by PVA/Chitosan Scaffolds in Canine Spinal Cord Injury Model

Deas Makalingga Emiri, Wilhelmina Wilma Wijaya

Emir Ryandika Samyudia, Yudha Mathan Sakti, Zikrina Abyanti Lanodiyu

BACKGROUND

Spinal Cord Injury



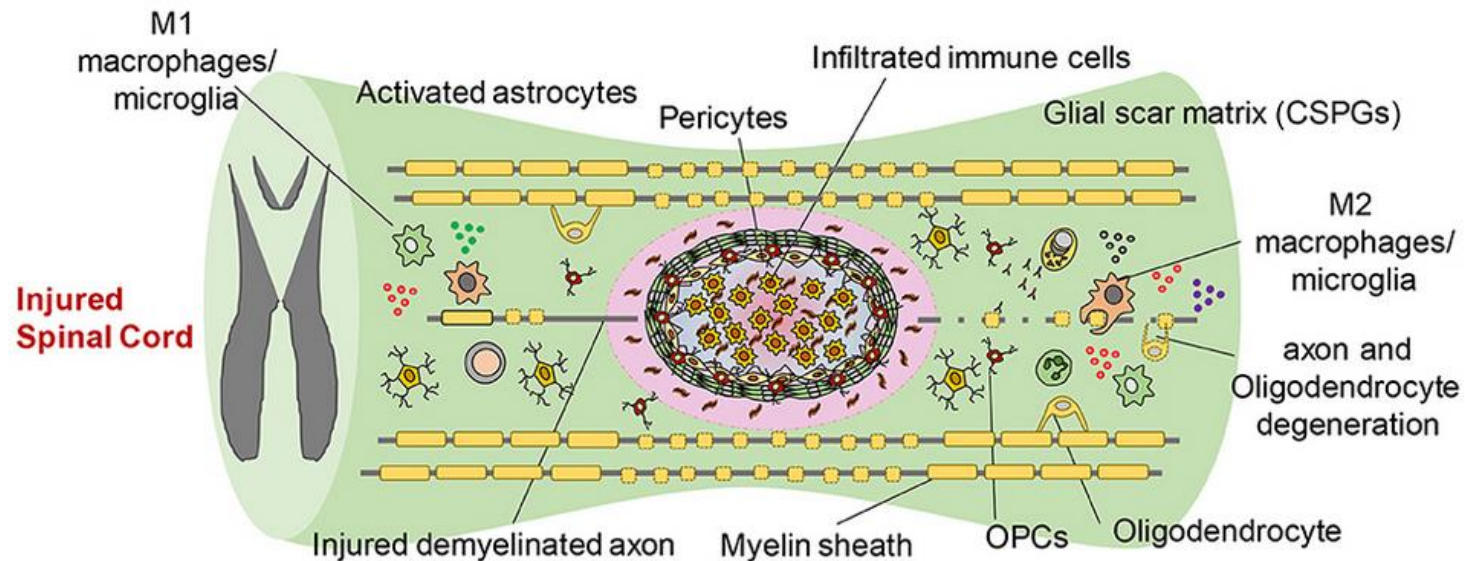
Cellular and vascular disruption



Necrosis, Necroptosis, and Apoptosis



Neuroinflammation



Glial scar formation around the lesion



Intramedullary Cyst Formation

Tissue damage



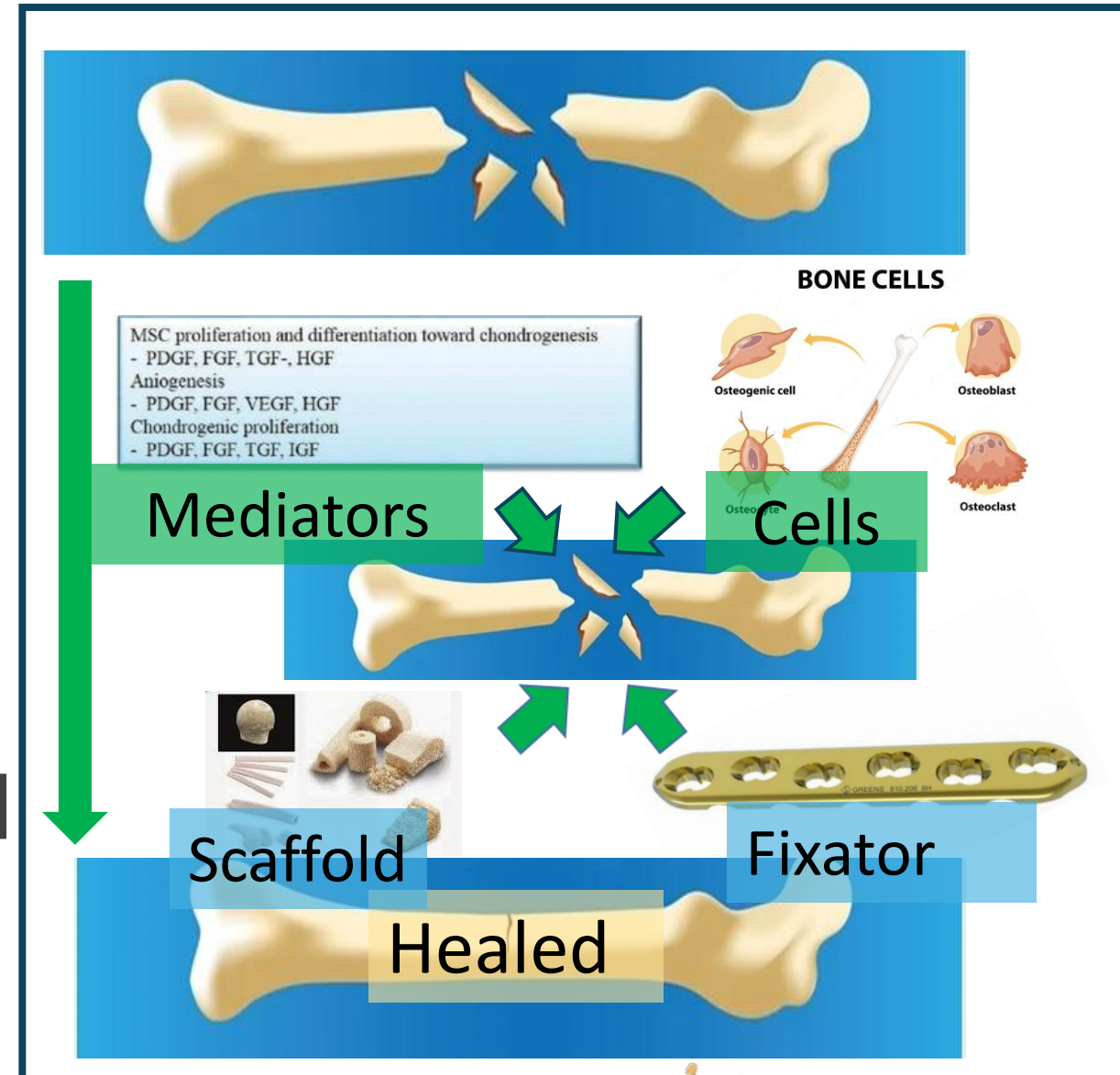
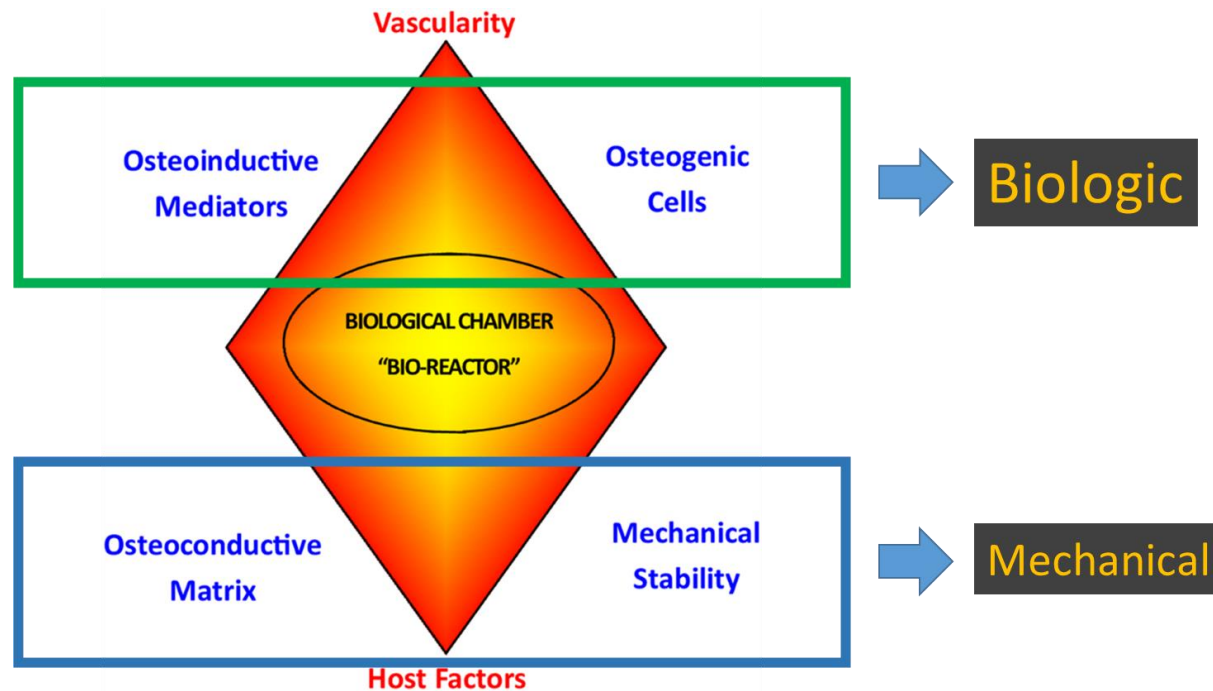
Structural + Functional Gap

Main Therapeutical Challenge



BACKGROUND

'Diamond Concept' in Orthopaedic



BACKGROUND

Bone Healing

Osteogenic

Osteoinductive

Osteoconductive

Mechanical Stability

Adopting 'Diamond Concept' from bone healing into neuron regeneration for better outcome

Neuron Regeneration

Cellular Platform

Neurinductive

Neuroconductive

Structural Integrity



BACKGROUND

Biological Aspect

Cellular Platform

- Hypoxia and cellular damage causing excitotoxic condition. Leading to cell death
- **Functional gap**

Neuroinductivity

- **ROS Formation** and inflammation process
- Regenerative inhibitory factors

Diamond Concept In Spinal Cord Injury

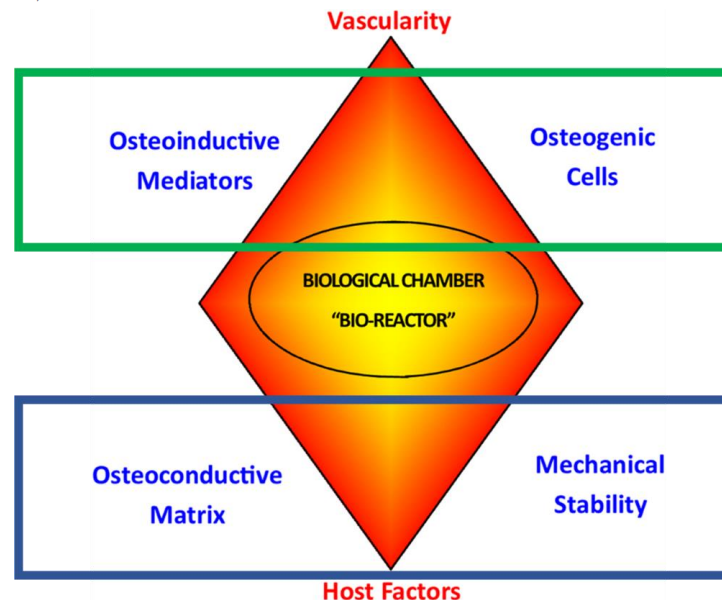
Scientific Foundation SPIROSKI, Skopje, Republic of Macedonia
Open Access Macedonian Journal of Medical Sciences. 2021 Nov 29; 9(F):754-769.
<https://doi.org/10.3889/oamjms.2021.7438>
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Section: Narrative Review Article



Diamond Concept as Principle for the Development of Spinal Cord Scaffold: A Literature Review

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Mechanical Aspect

Neuroconductivity

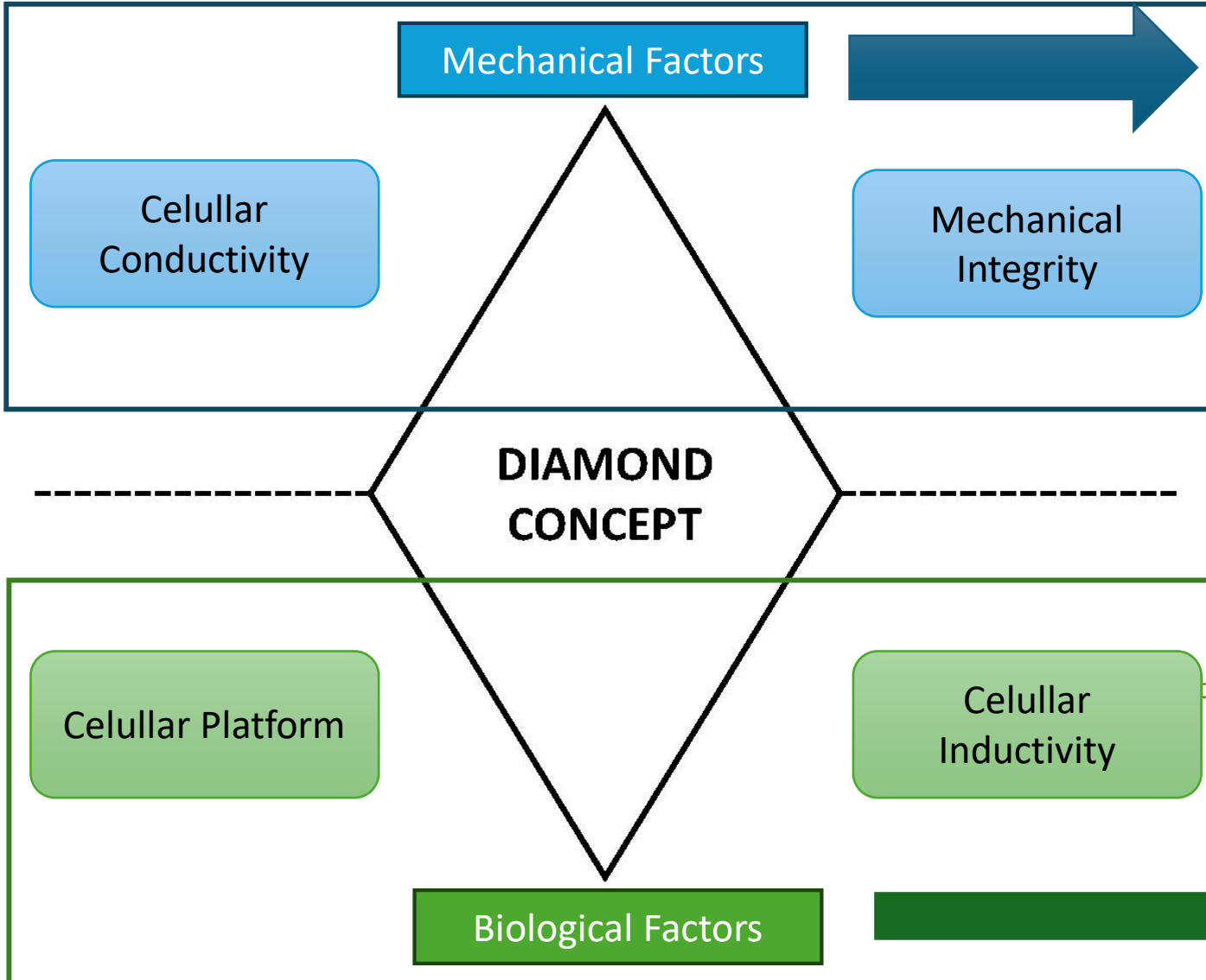
- Cystic formation due progressive tissue death
- **Structural gap**

Structural Integrity

- Instability due to structural damage surround the spinal cord
- Causing extent of tissue damage



BACKGROUND



Introductory Article



A Clinical Practice Guideline for the Management of Acute Spinal Cord Injury: Introduction, Rationale, and Scope

Michael G. Fehlings, MD, PhD, FRCS, FACS^{1,2}, Lindsay A. Tetreault, PhD^{1,3}, Jefferson R. Wilson, MD, PhD^{2,4}, Brian K. Kwon, MD, PhD⁵, Anthony S. Burns, MD, PhD², Allan R. Martin, MD¹, Gregory Hawryluk, MD, PhD⁶, and James S. Harrop, MD⁷



Current Guideline Based Therapy

Healing

The Journal of Spinal Cord Medicine



J Spinal Cord Med. 2009 Apr; 32(2): 105–114.
doi: [10.1080/10790268.2009.11760761](https://doi.org/10.1080/10790268.2009.11760761)

PMCID: PMC2678281
PMID: [19569457](https://pubmed.ncbi.nlm.nih.gov/19569457/)

Stem Cell-Based Therapies for Spinal Cord Injury

[Rishi S. Nandoe Tewarie](#), MD,^{1,2} [Andres Hurtado](#), MD,² [Ronald H Bartels](#), MD, PhD,¹ [Andre Grotenhuis](#), MD, PhD,¹ and [Martin Oudega](#), PhD^{2,3}

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Regenerative medicine with the use of stem cells and derivatives is expected to revolutionize patient treatment, answering the unanswered questions

Fehlings, M. G., Tetreault, L. A., Wilson, J. R., Kwon, B. K., Burns, A. S., Martin, A. R., Hawryluk, G., & Harrop, J. S. (2017). A Clinical Practice Guideline for the Management of Acute Spinal Cord Injury: Introduction, Rationale, and Scope. *Global Spine Journal*, 7(3_suppl), 84S-94S. <https://doi.org/10.1177/2192568217703387>



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BACKGROUND

Stem cell alone is not enough

Each cell needs a certain **supportive microenvironment** to support the **activity**.

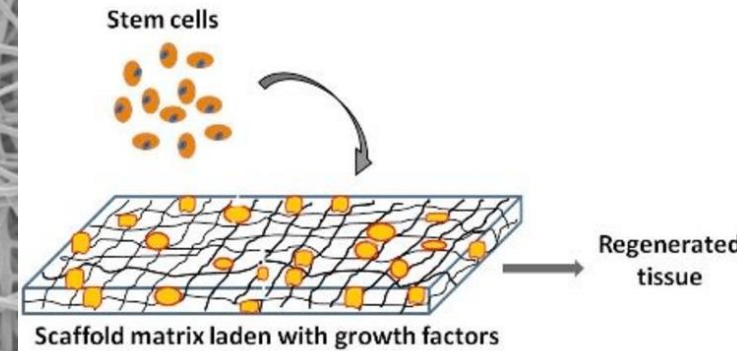
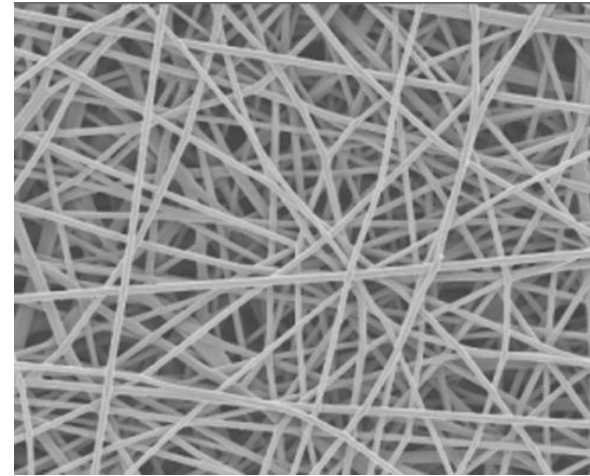


To **differentiate** into a mature and specific cell



Able to facilitate cell **migration and adhesion**.

Without host cellular development, any intervention won't be able to support regeneration process

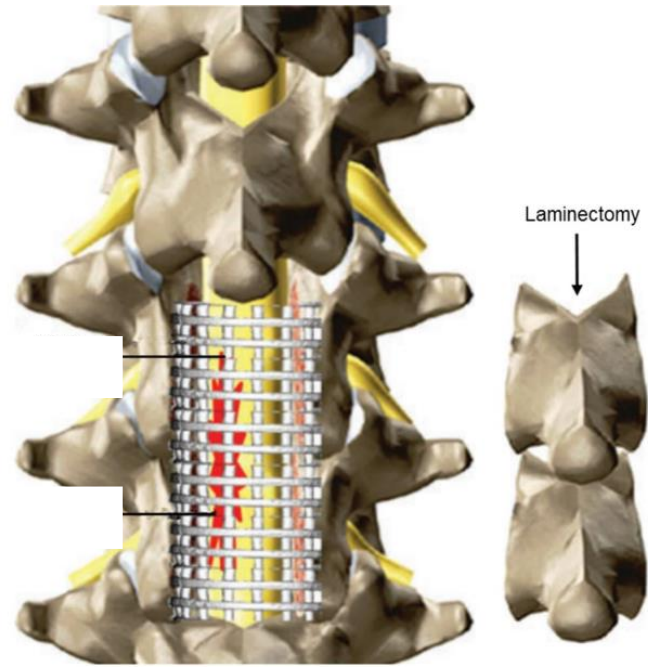


PVA/Chitosan revealed as promising candidate with expected biological and structural properties



BACKGROUND

What is the regenerative capability of combining a nanofibrous PVA/Chitosan scaffold with allogeneic umbilical cord-derived stem cells in a canine SCI model?



How is the **clinical improvement** in the canine spinal cord injury model treated with combination of PVA/Chitosan and allogeneic Umbilical Cord mesenchymal stem cells?

How is the **severity of Intra-lesion Hemorrhage and spinal cord inflammation** in the canine spinal cord injury model treated with combination of PVA/Chitosan and allogeneic Umbilical Cord mesenchymal stem cells?

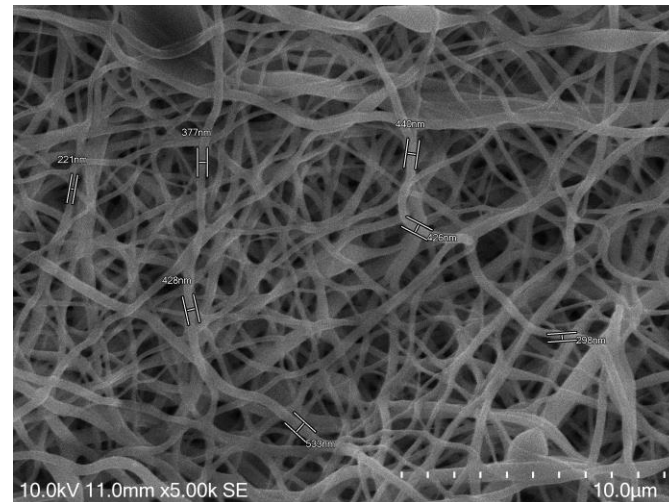
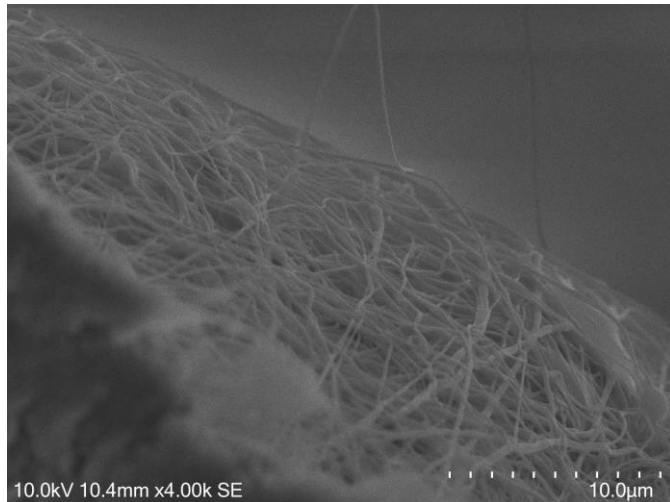
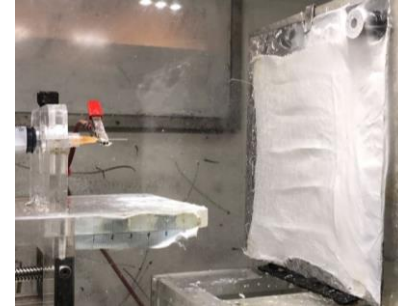
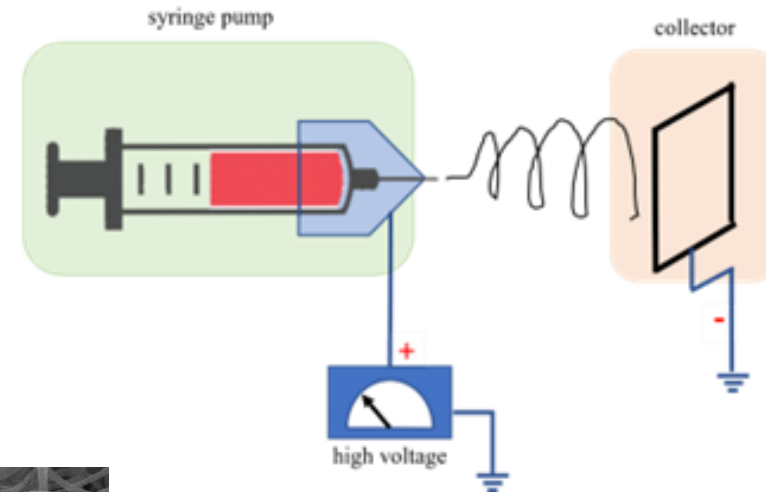
How is the **regeneration capability** of combination of PVA/Chitosan and allogeneic Umbilical Cord mesenchymal stem cells in treating canine spinal cord injury model?



METHODS

Production of PVA/Chitosan Scaffold

1. PVA and Chitosan material mixed with the composition 85:15 (PVA:Chi) creating a liquid phase composite
2. Liquid phase composite prepared inside 10 cc syringe with needle sized 0.4 mm and collecting distance is 14 mm
3. Needle is electrostatically charged to perform electrospinning process, resulting nanosized fiber



Measurement	Result (6 ml)
Scaffold Thickness	$163.66 \pm 4.37 \mu\text{m}$
Porosity	$19.31 \pm 1.41 \%$
Fiber Width	$460.25 \pm 26.84 \text{ nm}$



METHODS

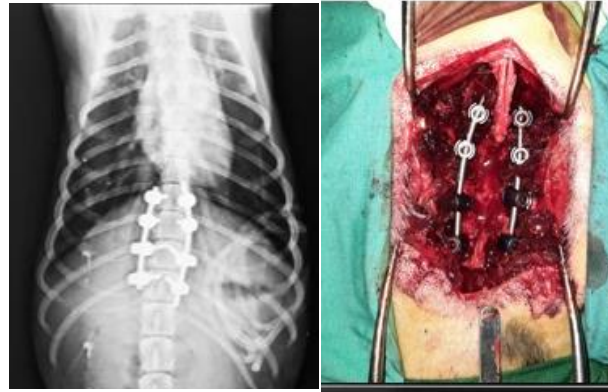
Control Group

Balloon compression
method spinal cord
injury model



Instrumentation

Laminectomy,
Decompression and
Stabilization



Combination

Instrumentation, Scaffold
implantation within
duramater and stem cell
delivery



Subject Criteria

- Male Canis Lupus Familiaris
- Weight : 10-20 kg
- Healthy (Examination by veterinarian)
- No Motoric Dysfunction (cBBB = 19)

Paralysis Criteria

- cBBB Score = 0
- Patella Reflex = +3
- Hot Sensory Respond = > 30 seconds

JOURNAL of MEDICINE and LIFE

JML | ORIGINAL ARTICLE

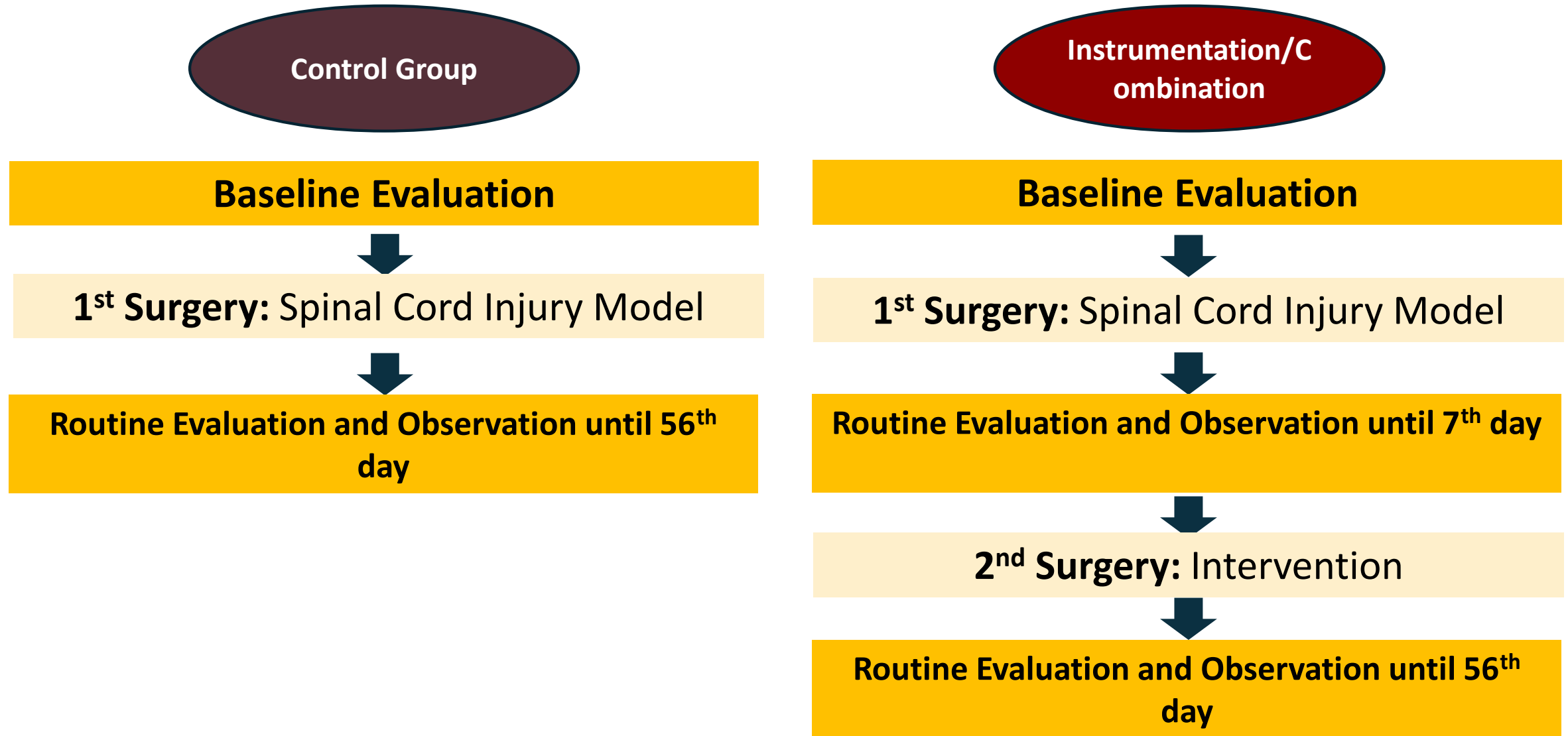
Balloon compression-induced spinal cord injury in canines: a large animal model for spinal cord injury research

Yudha Mathan Sakti¹, Emir Riandika Samyudia¹, Deas Makalingga Emiri¹, Teguh Aryandono², Rahadyan Magetsari¹, Rusdy Ghazali Malueka³, Ery Kus Dwianingsih⁴

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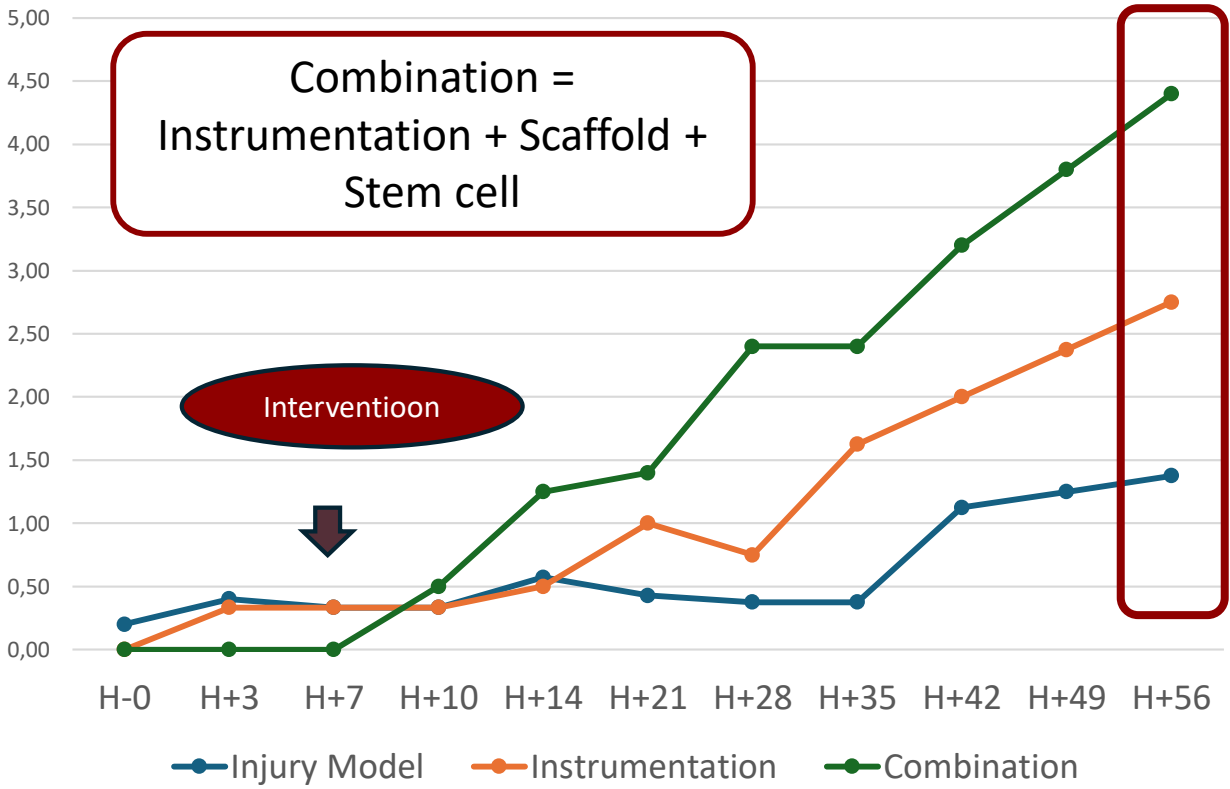


METHODS



RESULT : Clinical Improvement

Progression of cBBB score



Observer 1 : Emir Ryandika Samyudia
Observer 2 : Deas Makalingga Emiri

Nilai	Sig.	95% CI
0,784	,000	-
Jumlah Observasi N = 143		

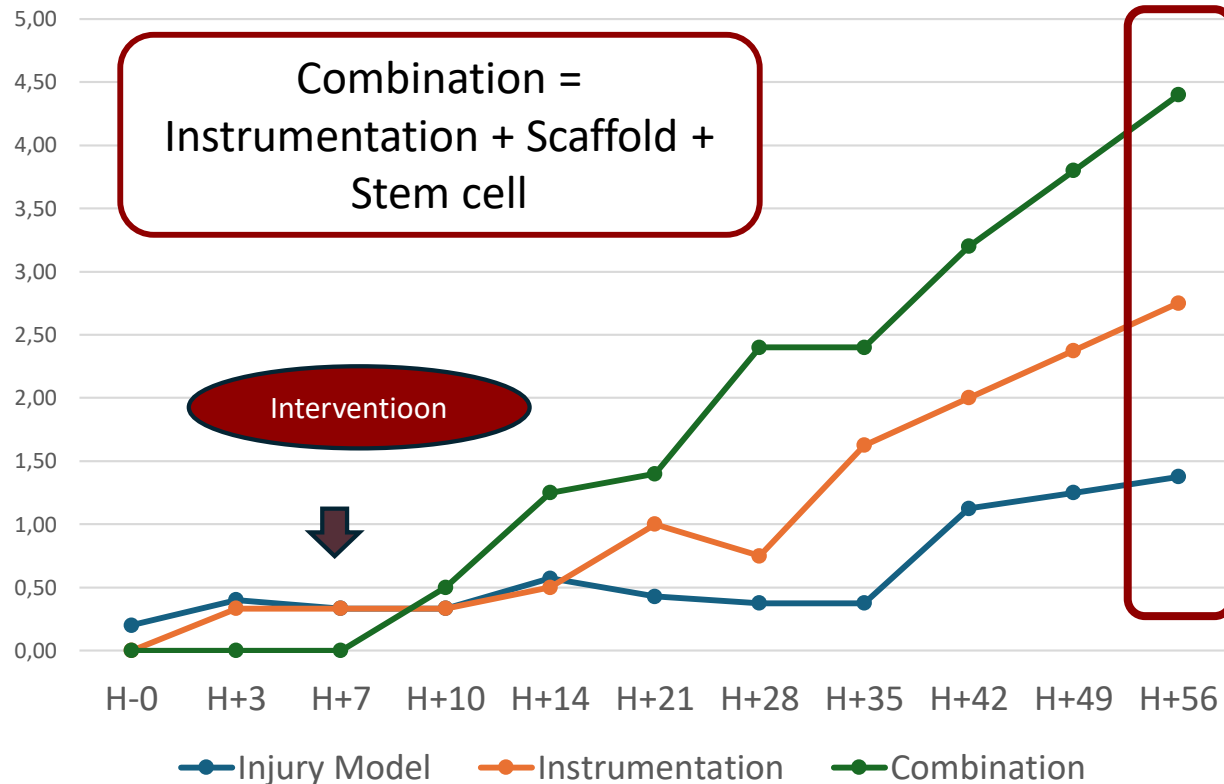
Inter Observer Reliability Test (2 Observer)

	H-0	H+3	H+7	H+10	H+14	H+21	H+28	H+35	H+42	H+49	H+56
CD	0,20	0,40	0,33	0,33	0,57	0,43	0,38	0,38	1,13	1,25	1,38
IM	0,00	0,33	0,33	0,33	0,50	1,00	0,75	1,63	2,00	2,38	2,75
SC	0,00	0,00	0,00	0,50	1,25	1,40	2,40	2,40	3,20	3,80	4,40

Daily Average cBBB Score

RESULT : Clinical Improvement

Progression of cBBB score



Observer 1 : Emir Ryandika Samyudia
Observer 2 : Deas Makalingga Emiri

Comparison	Avg.	Std. Deviation	Sig.
Injury Model - Instrumentation	.03667	.61426	.840
Instrumentation – Combination	-.79833	.92229	.037
Injury Model – Combination	.76167	1.11071	.012

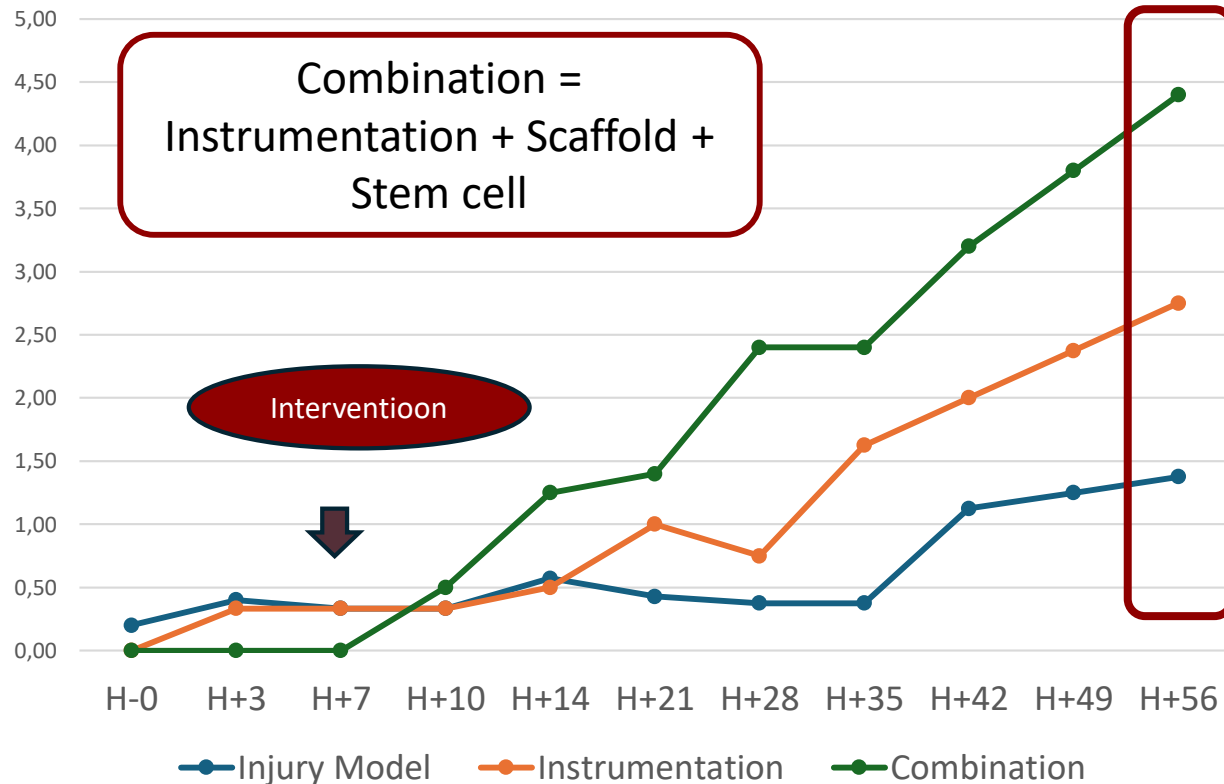
Statistical Analysis (Mann Whitney – Pairwise Comparison)

Result : **Motoric function** were **significantly improved** in the combination group compared to other groups



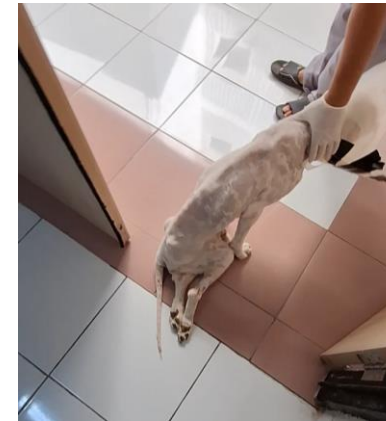
RESULT : Clinical Improvement

Progression of cBBB score



Observer 1 : Emir Ryandika Samyudia
Observer 2 : Deas Makalingga Emiri

Subject Footage



Model Cedera



Implan



Kombinasi

Significant cBBB score improvement

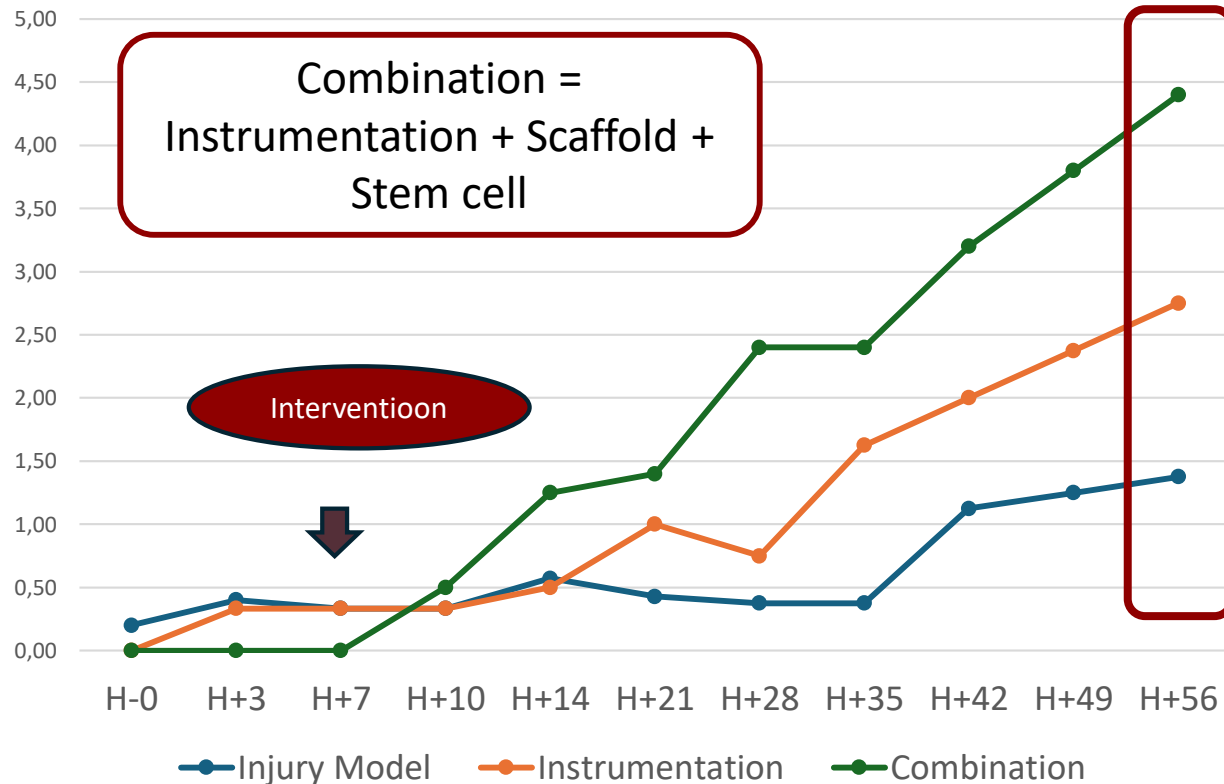


Significant observed motoric function



RESULT : Clinical Improvement

Progression of cBBB score



Observer 1 : Emir Ryandika Samyudia
Observer 2 : Deas Makalingga Emiri

Delivery stem cells in the acute phase will inhibit debris clearance activity of microglia and macrophages in tissue cleansing
(Xia et al., 2023)

Inhibited debris clearance could cause neuronal apoptosis

Not Ideal

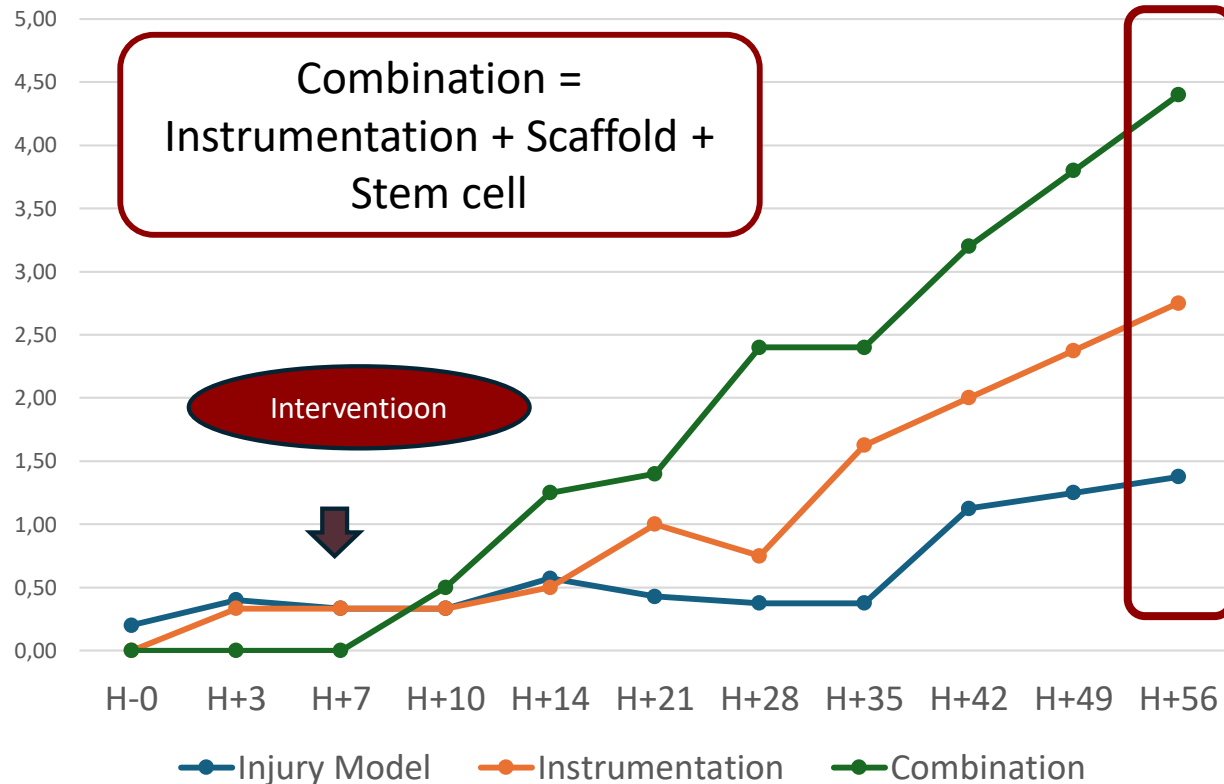
Each groups show improvement when entered subacute phase ~ 7 days

Subacute ideal to promote neuro-regeneration



RESULT : Clinical Improvement

Progression of cBBB score



Observer 1 : Emir Ryandika Samyudia
Observer 2 : Deas Makalingga Emiri

Mesenchymal stem cells provide neural repair by inducing sequential microglial activation after central nervous system injury

Mesenchymal stem cells increased cytokine expression that promoted effective regeneration of M2 at subacute phase after central nervous system injury
(Qu and Zhang, 2017; Xin et al., 2023)

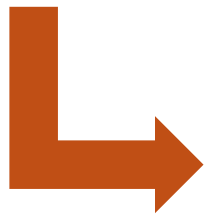
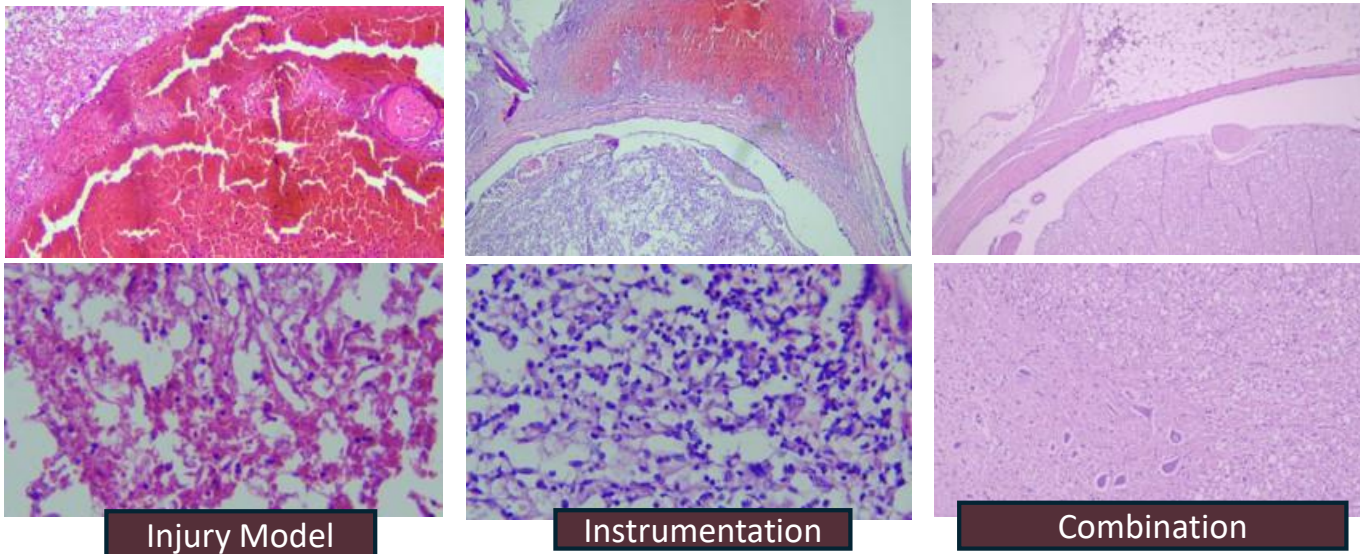
Mendukung hasil perbaikan fungsi motorik pada kelompok kombinasi sel punca secara signifikan dibandingkan 2 kelompok lain mulai hari ke-10

Result : The combination of allogeneic **stem cells** from umbilical cord and PVA/Chitosan composite **scaffold** has **the regenerative ability in restoring motoric function**



RESULT : Intra-lesion

Hemorrhage Anatomical Pathology (HE)



Observationally, more bleeding was observed in the injury model group compared to the other 2 groups

Combination =
Instrumentation + Scaffold +
Stem cell

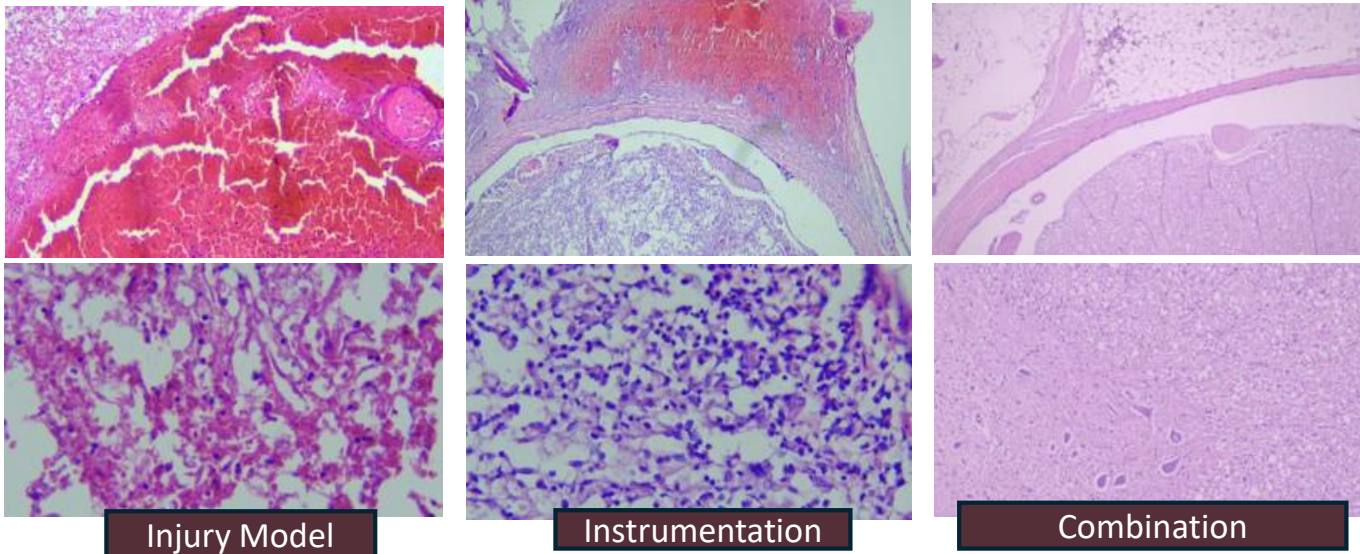
Observation	Pairwise Comparison	Sig.
Intralesion Bleeding	Combination - Instrumentation	0.580
	Combination – Injury Model	0.019
	Instrumentation – Injury model	0.045

Result : Intralesional bleeding of the Injury Model was significantly higher than that of the other groups



RESULT : Intra-lesion

He Anatomical Pathology (HE)



Another study obtained similar results in research using mesenchymal stem cells on mouse subjects, with the finding of more significant hemorrhage on injury model subjects (Menezes et al., 2020)

Implant instrumentation has the effect of controlling mechanical injury. Biological interventions of stem cells and scaffolds provide control of inflammation and hemorrhage at the injury site

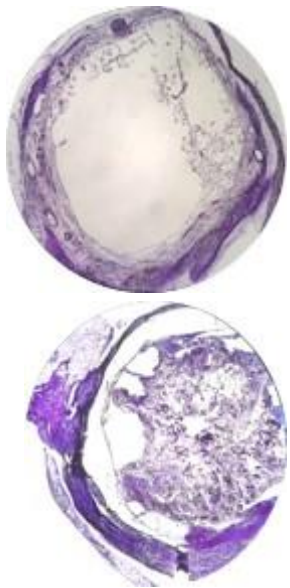
Observationally, more bleeding was observed in the injury model group compared to the other 2 groups

Result : The combination of allogeneic stem cells from umbilical cord and PVA/Chitosan composite scaffold has the ability to reduce the severity of hemorrhage



RESULT : Regeneration

Capability Anatomical Pathology (Luxol Fast blue)

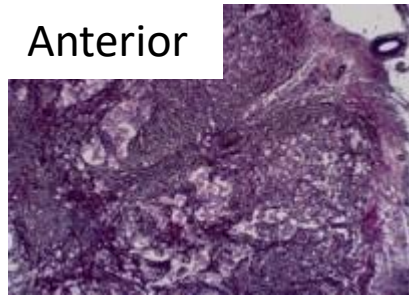


Injury Model

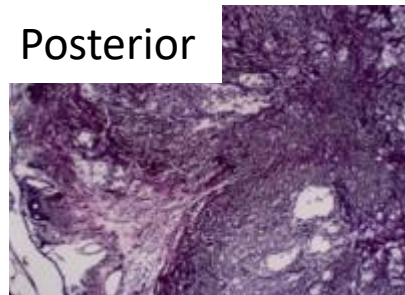


Damage in
almost entire
tissue

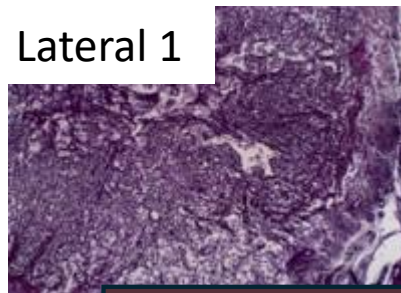
Anterior



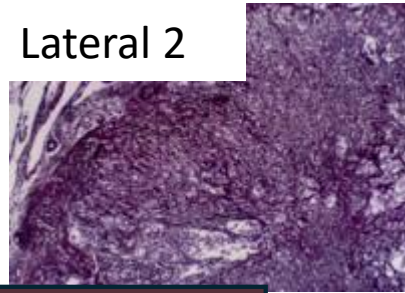
Posterior



Lateral 1



Lateral 2

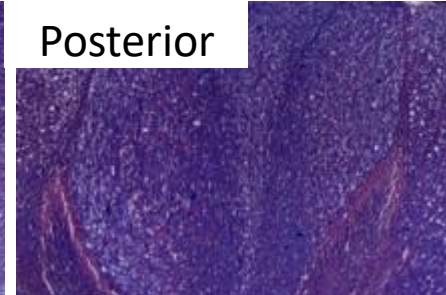


Instrumentation

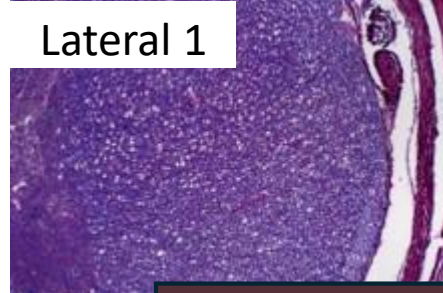
Anterior



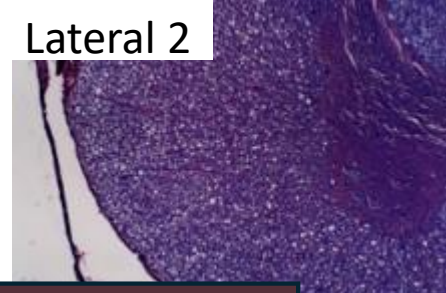
Posterior



Lateral 1



Lateral 2



Combination

Result: The condition of the myelin sheath was **better** in the **combination group** compared to the other 2 groups



RESULT : Regeneration

Capability Anatomical Pathology (Luxol Fast blue)

Focus of Observation		Sig.
Intra-Lesion	Anterior	0.028
	Posterior	0.100
	Lateral 1	0.034
	Lateral 2	0.018

Pairwise
Comparison



Observation	Inter-Group	Anterior Sig.	Lateral 1 Sig.	Lateral 2 Sig.
Intra-lesi	Combination - Instrumentation	0.020	0.048	0.027
	Combination – Injury Model	0.016	0.012	0.007
	Instrumentation – Injury model	0.840	0.499	0.513

The significant findings in the luxol fast blue results is well correlated with the results found in clinical motor functional testing

Reducing tissue damage and the demyelination process will directly increase the regeneration process capability. (Ahuja *et al.*, 2017)



Results: The combination of allogeneic stem cells from the umbilical cord and PVA/Chitosan composite scaffold has the ability to reduce demyelination after spinal cord injury



DISCUSSION

Synthesis of Nanofiber Membrane Scaffold

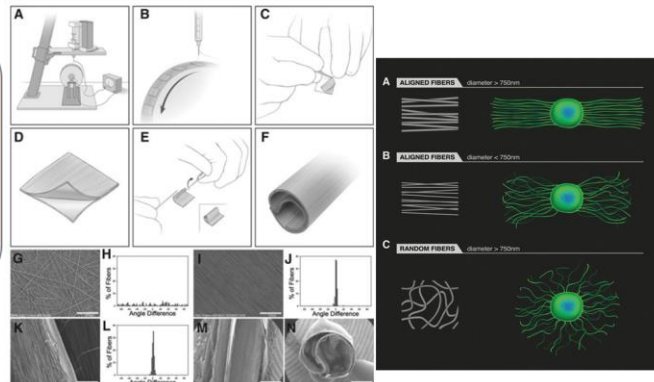
Currently, scaffolding is being develop in the treatment of spinal cord injuries

Scaffold proven to be able to assist remyelination or regeneration the axons and to integrate new neural cells in the SCI

scaffold → to **bridge the gap** of the lesion as **contact guidance for axonal growth**, act as a vehicle to deliver stem cells

- Electrospinning is promising option in producing the scaffold.

Electrospun fiber scaffolds with greater fiber alignment result in a more elongated DRG (dorsal root ganglia) body, with longer neurites that align to the orientation of fibers compared with results where DRG were cultured on randomly organize fibers.



Schaub NJ, Johnson CD, Cooper B, Gilbert RJ. Electrospun fibers for spinal cord injury research and regeneration. *Journal of neurotrauma*. 2016 Aug 1;33(15):1405-15.



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[J Neurotrauma](#). 2016 Aug 1; 33(15): 1405–1415.

doi: [10.1089/neu.2015.4165](https://doi.org/10.1089/neu.2015.4165)

PMCID: PMC4971409

PMID: [26650778](https://pubmed.ncbi.nlm.nih.gov/26650778/)

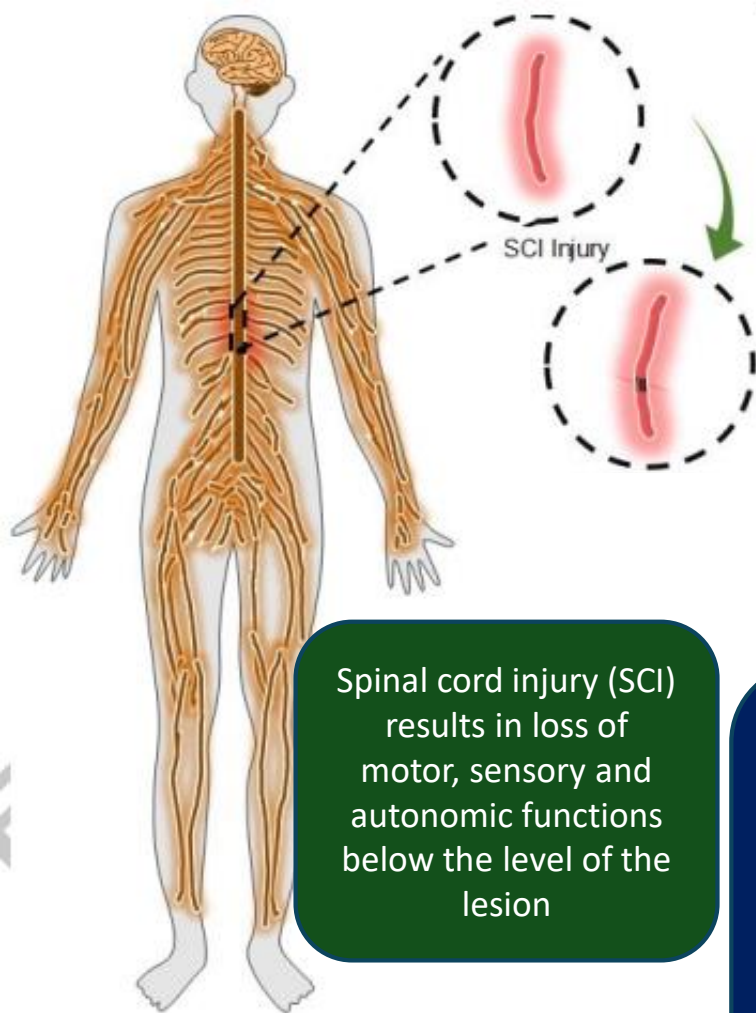
Electrospun Fibers for Spinal Cord Injury Research and Regeneration

[Nicholas J. Schaub](#),^{1,2,*} [Christopher D. Johnson](#),^{1,2} [Blair Cooper](#),³ and [Ryan J. Gilbert](#)^{1,2}

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Current Therapy



Treatment modalities

- 1 Spinal fixation
- 2 Drugs
- 3 Neurotropic agent
- 4 Stem cells only

Limited use and high side effect!

Downside

- Risk of in-situ volume compression due to unattached cell in stem cell only therapy
- Most Drugs unable to transport through blood brain barrier
- No control release of drug

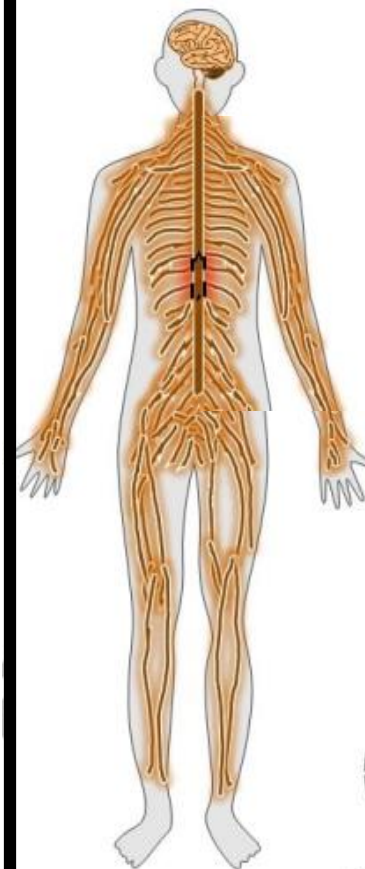
Spinal cord injury (SCI) results in loss of motor, sensory and autonomic functions below the level of the lesion

Promising Therapeutical Option

PVA-Chi Spinal Scaffold

Nanosized fiber favoring cell adhesion

Non-Toxic for Cells. Able to be Combined with cellular therapy

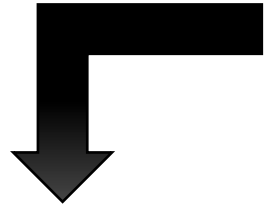


PVA-Chi and UC-MSC

Enhance the spinal cord regeneration after the injury



DISCUSSION



Autologous

- Safer immune and genetic profile
 - Low risk of rejection
- Causing pain in harvest process and risk of iatrogenic injury
 - Limited volume
 - Takes time

Allogenic

- Need donor selection
 - Higher risk of rejection
- No risk of iatrogenic injury
 - On Demand Preparation
 - Various cell availability



Autologous versus allogenic mesenchymal stem cell therapy: The pros and cons

Nisha Durand   • Abba C. Zubair
Published: December 01, 2021 •

Autologous has a **low risk of rejection** but allogenic has advantages in **preparation** and **flexibility in the type of cell source**



DISCUSSION

Comparison of Mesenchymal Stem Cell (MSC) Sources and Cellular Characteristics

Growth Potential

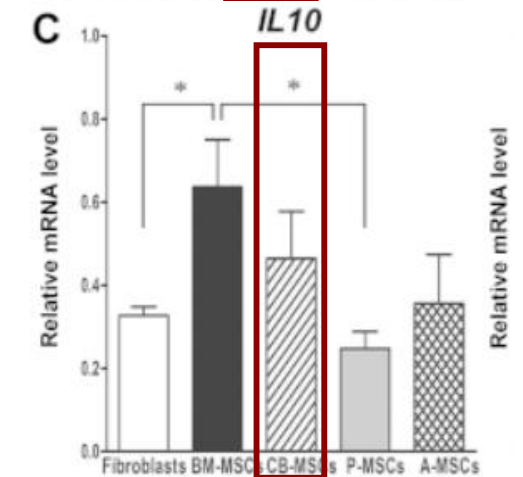
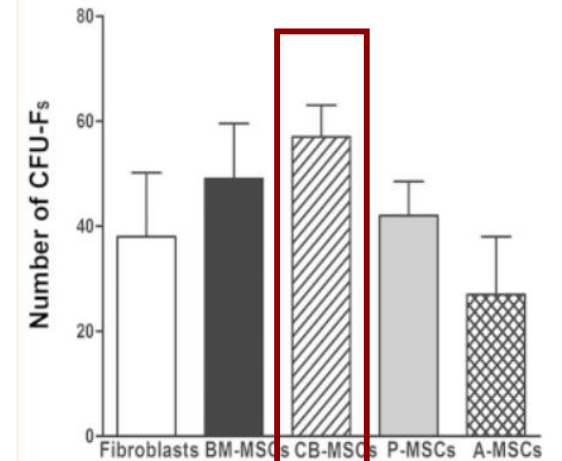
- **Umbilical Cord** Mesenchymal Stem Cell has good proliferation capacity
- **Adipose** Mesenchymal Stem Cell has less proliferation capacity compared to other

Immunosuppression

- **Bone marrow** Mesenchymal Stem Cell has the most potent immunosuppressive characteristic, followed by **Umbilical cord** source

Comparison of molecular profiles of human mesenchymal stem cells derived from bone marrow, umbilical cord blood, placenta and adipose tissue

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DISCUSSION

Comparison of Mesenchymal Stem Cell (MSC) Sources and Cellular Characteristics

Growth Potential

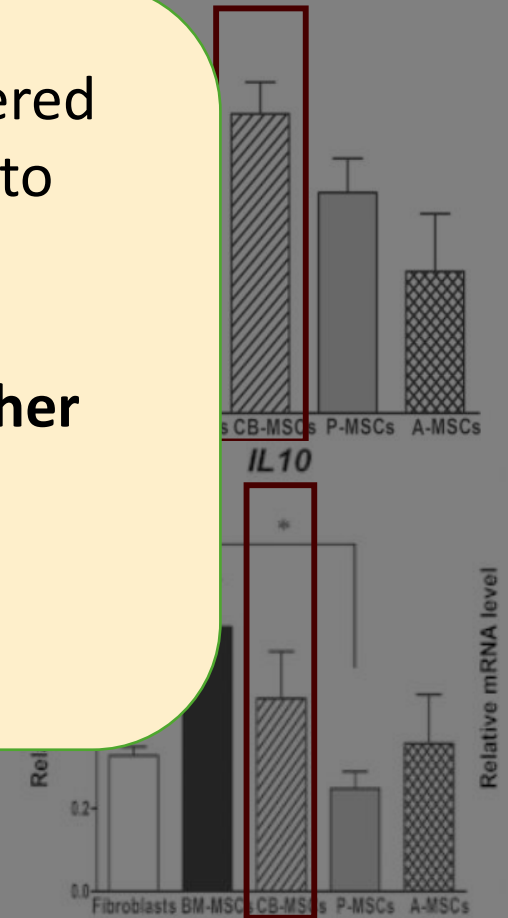
Umbilical Cord Mesenchymal Stem Cell considered ideal for cell based regenerative therapy due to

1. Ease of tissue harvesting
2. has a better growth potential compared to other source
3. adequate immunosuppression capacity
4. facilitates regeneration

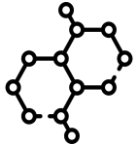
Immunosuppression

Comparison of molecular profiles of human mesenchymal stem cells derived from bone marrow, umbilical cord blood, placenta and adipose tissue

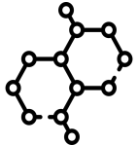
Open Access



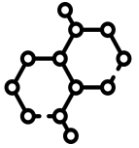
Conclusion



The combination of allogeneic stem cells from umbilical cord and PVA/Chitosan composite scaffold has regenerative ability in restoring motoric function



The combination of allogeneic stem cells from umbilical cord and PVA/Chitosan composite scaffold has the ability to reduce the severity of hemorrhage



The combination of allogeneic stem cells from the umbilical cord and PVA/Chitosan composite scaffold has the ability to reduce demyelination after spinal cord injury



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Thank You